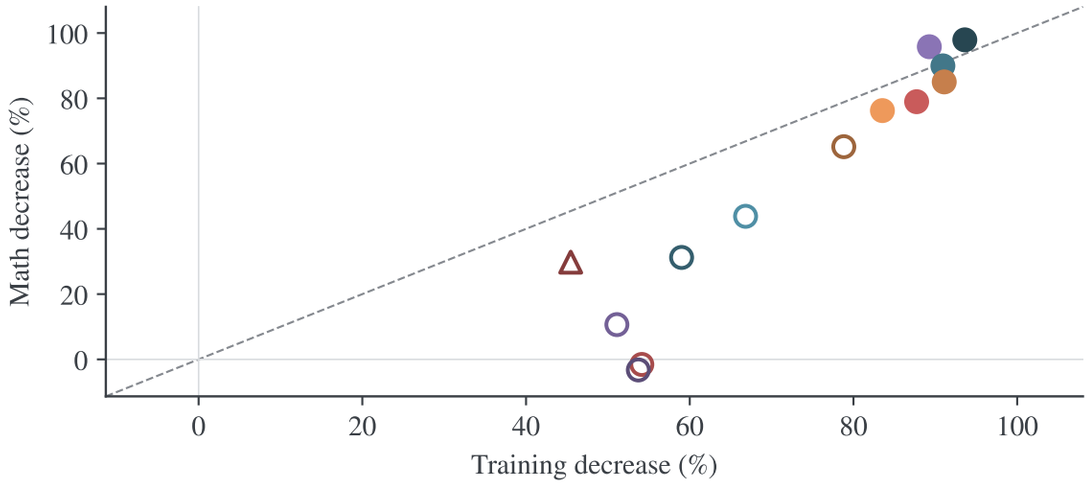
Qwen3-1.7B / thinking | Training pool vs. held-out

Relative RKL decrease (%)

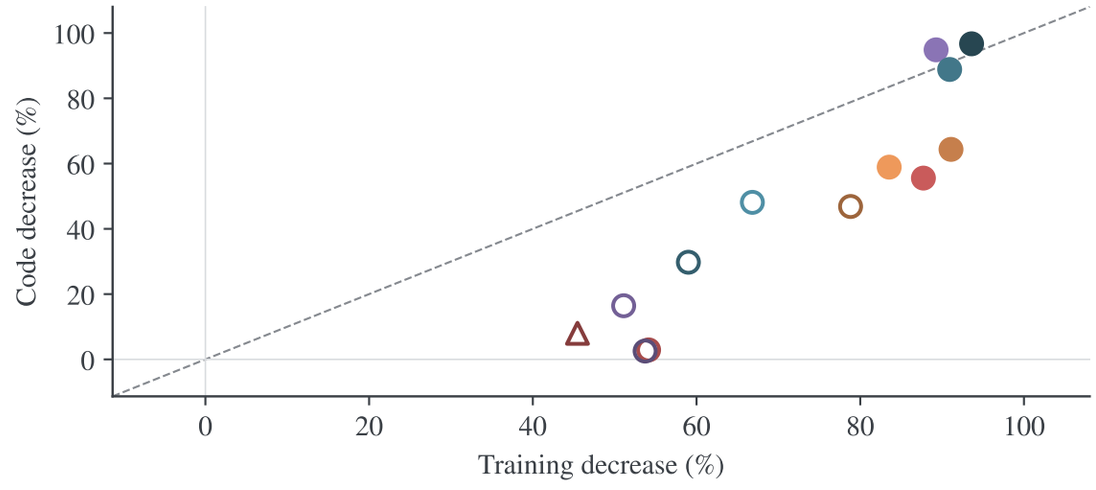
(a) Training vs. Math

n=13; r=0.94; rho=0.93



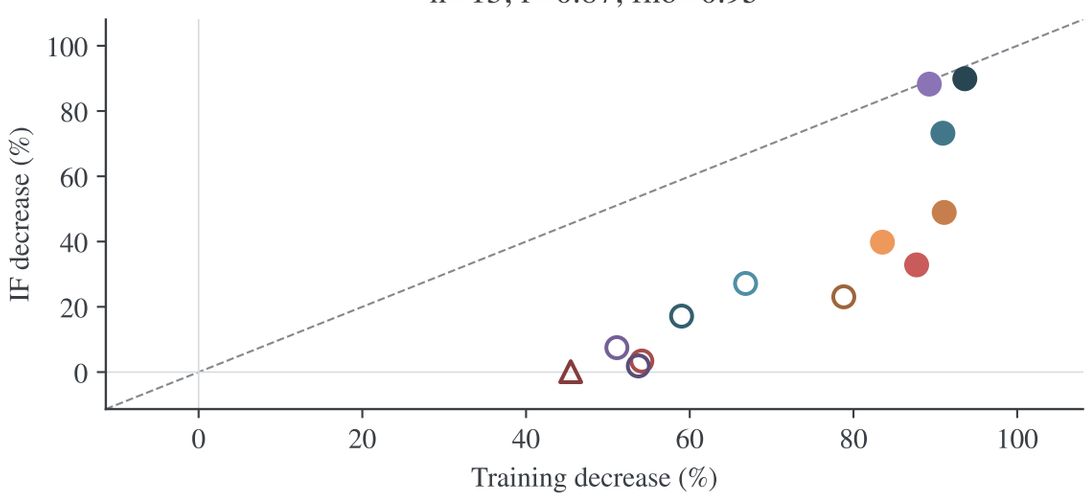
(b) Training vs. Code

n=13; r=0.93; rho=0.92



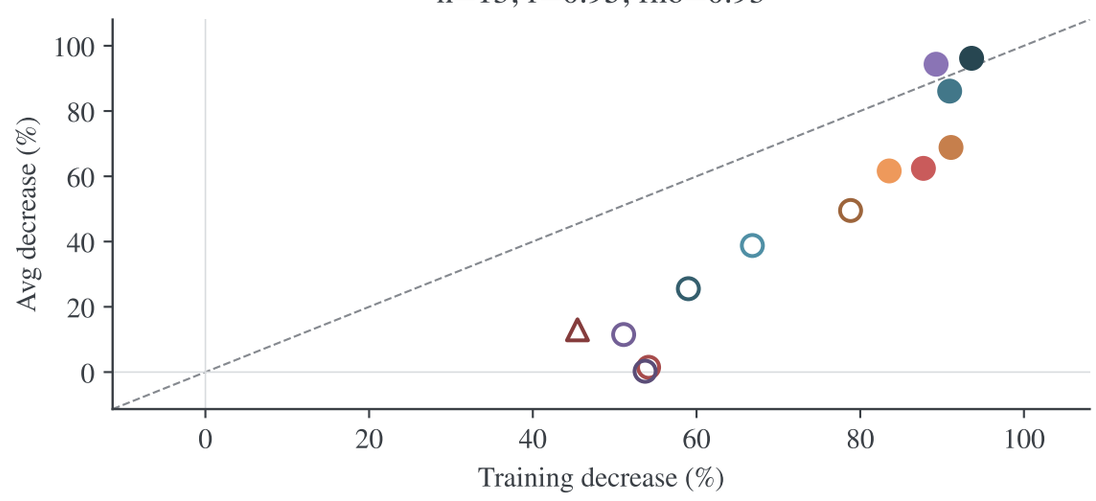
(c) Training vs. IF

n=13; r=0.87; rho=0.95



(d) Training vs. Avg

n=13; r=0.95; rho=0.93



- Qwen3-0.6B [320]
- Qwen3-1.7B-Base [320]
- Qwen3-1.7B-Base-OPD / Thinking-Space [320]
- Qwen3-1.7B-GRPO-KL-Math [320]
- Qwen3-1.7B-Math / Samsung SFT [320]
- Qwen3-1.7B-SFT / Thinking-Space [320]
- Qwen3-4B [320]
- Qwen3-4B Math-CoT SFT step640 [320]
- Qwen3-4B Math-CoT SFT step80 [320]
- Qwen3-4B-Base-GRPO / Thinking-Space [320]
- Qwen3-4B-Thinking-2507 [64]
- Qwen3-8B [320]
- e3-1.7B [320]
- Equal decrease

One point = one run endpoint; triangle: 64 updates; circle: 320; filled: same size; open: cross size.

Fixed-H RKL; dashed: equality; correlations pool all displayed endpoints (mixed budgets/sizes).